

Patient is a 38 year old female BW 55kg with Trisomy 21, is scheduled for dental rehabilitation and extractions under GA. She had PDA ligated in infancy and is on oral thyroxine for congenital hypothyroidism.

A. Outline your anaesthetic concerns in the management of this patient (50%)

(A)

My anaesthetic concerns are:

Patient

Trisomy 21

- Trisomy 21 is associated with systemic complications including airway abnormalities, aspiration risks, hypotonia, increased risk of cardiovascular disease, OSA, DM, cognitive impairment, epilepsy, leukemia, polycythaemia, poor immunity pre-disposing to recurrent infections.
- Patient should be seen in pre-op clinic for detailed assessment for these systemic involvement, with relevant labs ordered and reviewed, to facilitate planning for anaesthesia..
- Presence of URTI will necessitate postponement of surgery.

Previous PDA ligation

- Should not have impact on current surgery.
- Cardiology notes and recent TTE scans should be reviewed for other concomitant cardiac abnormalities, and managed as appropriate.

Possible obesity

- Trisomy 21 is associated with obesity.
- BW 55kg, but height was not stated.
- Increased risks of OSA, OHS, increased sensitivity to opioid (respiratory depression).
- LA infiltration by surgeon will reduce requirement for long acting opioids.

Congenital hypothyroidism

- Important that patient is in euthyroid state.
- To check pre-op TFT.
- Risk of myxoedema coma, if patient is hypothyroid.

Poor dentition (indication for dental rehab)

- Increased risks of dental trauma during airway manipulation, especially if patient has pre-existing difficult airway features.

Anaesthesia

Airway concerns

- Trisomy 21 is associated with potential difficult intubation and aspiration risks.
- Potential difficult intubation (micrognathia, high arch palate, subglottic stenosis) which require video laryngoscope, airway adjuncts such as stylet/ bougie/ fibreoptic scope, and downsizing ETT size,
- Bag mask ventilation may be challenging due to poor mask fitting (midface hypoplasia, flattened nasal bridge) and upper airway obstruction (from macroglossia, enlarged tonsils/ adenoids).
- May have atlantoaxial subluxation – lateral flexion-extension neck CXR, minimal neck manipulation during intubation if there are any concerns.
- Increased aspiration risk due to delayed gastric emptying (may need prolonged fasting, or pre-op US gastric scan).
- If patient has obvious reflux symptoms on medications, RSI will be required.

IV access may be challenging due to obesity/ thin skin/ small veins

- US may be helpful to improve success rates.

Mental capacity/ inability to cooperate

- Trisomy 21 is associated with developmental delay, microcephaly, and mental retardation.
- Request for GA may be patient's preference, or it may reflect issues with mental capacity/ ability to cooperate.

- May need surrogate/ consultant consent.
- Premedication, parental presence may be needed if patient is unable to cooperate with IV cannulation.
- Patient may not be able to express pain verbally, and adequate analgesia should be given to ensure patient's comfort.

Surgical

Choice of airway

- Check with surgeon if nasal (vs oral) intubation is preferred to facilitate surgical access
- Check if throat pack is required by surgeon. Care should be taken to remove throat pack prior to extubation, with use of visual cues (label stickers) and systemic reminder/ prompts as required.

Shared Airway

- Risk of tube dislodgement/ kink/ migration.
- Close monitoring of airway pressure, capnography important for early detection.
- Care must be taken to secure tube.